

SANKUR KUNDU

Software Engineer | Backend, Systems & Full-Stack

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SUMMARY

Software engineer (B.Tech CSE, 2026) who builds systems from scratch in Rust and proves them correct: **tessel** (GPU/TPU kernel compiler: **cuBLAS-parity** GEMM, LLM engine at **1.05× vLLM**), **kiln** (an ML compiler), **ferrolm** (an LLM server), **quorumdb** (distributed SQL, **TLA+**-verified) and **faultline** (Jepsen-style tests of **etcd**). Co-founded Gigzen as its sole engineer.

PROJECTS

tessel — Rust, zero dependencies: a tile language and compiler for GPUs and TPUs

- Designed a **Triton-style tile language** and wrote its compiler from scratch: typed tile IR, **linear layouts**, **CUDA** for **tensor cores** (ldmatrix, mma.sync, swizzled shared memory, pipelining), an **auto-tuner**, plus **Metal** and **TPU (Pallas)** backends.
- On an **NVIDIA T4**: GEMM at **parity with cuBLAS**, attention **1.3–1.9× faster than PyTorch SDPA**; an LLM engine on its kernels matches **vLLM's** output token for token at **1.05× its throughput** (32 sequences).

kiln — Rust, zero dependencies: an ML compiler, ONNX to SIMD and CUDA kernels

- Built from scratch an **ML compiler** (XLA/TVM-style): **ONNX** import, **operator fusion**, an affine **loop IR**, **SIMD** and **CUDA** code generation (**tensor cores**, **CUDA** graphs), **auto-tuning** and memory planning; output matches **PyTorch**.
- On 4 CPU cores, **faster than torch.compile**, **ONNX Runtime** and **JAX/XLA**: **BERT 117 ms** vs 149, 153, 189. On an **NVIDIA T4** in fp16, **1.65× faster than torch.compile** on SmolLM2 (**7.9 ms** vs 13.1) and **3.7×** on an MLP.

ferrolm — Rust, zero dependencies: an LLM inference server and RAG stack

- Built from scratch an **OpenAI-compatible LLM server** with a **paged KV cache**, **continuous batching**, **prefix caching** and **speculative decoding** on hand-written **SIMD kernels**; runs SmolLM2 and Qwen2.5, matching **Hugging Face transformers**.
- On 4 CPU cores: **3.4× throughput** (continuous batching), first token **46× sooner** (prefix caching), **20× more requests** in the same memory (paging), **1.47× faster** decoding at identical output (speculation).

quorumdb — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built from scratch a **distributed SQL database** that **psql** connects to: **LSM storage**, **Raft**, a self-rebalancing **multi-Raft** store, **serializable ACID transactions** (Percolator), a SQL planner; **1,500 tpmC** on a **TPC-C**-derived benchmark.
- Proved by **deterministic simulation** (**10,000 Raft clusters**, **610K crashes**, histories machine-checked) and a **TLA+** model (**337K states**); found **6 real bugs**, catches all **15 planted**; **time-travel debugger** rewinds runs by replay.

faultline — Rust, zero dependencies: Jepsen-style fault injection for etcd

- Runs real **etcd** clusters in **Linux network namespaces**, injects **iptables partitions**, **kill -9** crashes and **SIGSTOP** pauses, and checks histories for **linearizability**, **watch completeness** and **lost updates**.
- Tested **4 etcd release lines** with **1.26M operations** under **288 faults**: linearizability and watches held; traced **lost updates** under etcd locks to a paused leader applying a timed-out write **6 s** late, and showed **fenced writes** prevent it.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer

Apr 2026 – Present

Bhubaneswar, India · *Early-stage startup (incorporation in progress)* · *Sole engineer across every product*

- Sole engineer** on **Waggle**, a delivery platform of **three Android apps** (riders, shops, customers) from one **TypeScript / React / Capacitor** codebase on **Supabase PostgreSQL**: **53 tables**, **90 RLS policies**, **154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: **first-accept-wins dispatch** (45-second offer window), **4-digit handover codes**, **Razorpay/UPI** payments reconciled by UTR, **GST invoices** and **pg_cron** billing runs.
- Security-tested** the platform as one system: **150 automated attacks** on the database access rules, all blocked; a **42-step Playwright** end-to-end run across all three apps, from rider verification to delivery rating.
- Built **Populace**, an open-source multi-user testing tool (**Node.js**, an **npm CLI**, **GitHub Action** and **Electron** app); its **first run found 5 defects in 3.5 minutes** after manual testing had passed, one breaking **every signup**.
- Designed its **13-method adapter contract** and proved it on third-party software (**Gitea**: 1,106 API calls); **3.17M API calls** across 30–300 simulated users with **zero API failures**; **122 self-tests** in CI.

SKILLS

CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems, DBMS internals, OS, Networks

Languages: Rust, TypeScript, JavaScript, Python, SQL (PL/pgSQL), Java

Backend & Data: PostgreSQL, Supabase, Row-Level Security, pg_cron, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI

Frontend & Mobile: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS

Systems & performance: CUDA (tensor cores), GPU/TPU kernel compilers, ML compilers, SIMD (AVX-512), LLM serving

Testing, DevOps & Cloud: Simulation testing, fault injection, TLA+, Playwright, GitHub Actions, Docker, Kubernetes, GCP

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata

Sep 2022 – Apr 2026